

RIDE_multi_app_simulated_data

September 11, 2026

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[1]: import random
import pandas as pd
from datetime import datetime, timedelta

# Configuration
NUM_RECORDS = 10000
APPS = ["Uber", "Ola", "Rapido", "Namma Yatri"]

data = []
start_time = datetime(2026, 8, 1, 8, 0, 0)

for i in range(1, NUM_RECORDS + 1):
    request_id = f"REQ{i:05d}"
    user_id = f"USR{random.randint(1000, 5000)}"

    # Simulate realistic booking timestamps
    request_time = start_time + timedelta(minutes=random.randint(1, 43200))

    # Broadcast across 2 to 4 apps simultaneously
    selected_apps = random.sample(APPS, k=random.randint(2, 4))
    connected_apps_str = ", ".join(selected_apps)

    # Determine initial driver match
    winning_app = random.choice(selected_apps)
    time_to_first_accept_sec = random.randint(12, 90)

    # System Auto-Cancel reliability check (e.g., API latency)
    auto_cancel_status = random.choices(
        ["Success", "Delayed_Fee_Incurred"],
        weights=[0.94, 0.06]
    )[0]

    # Driver Cancellation & Auto-Reassignment logic
    captain_cancelled = random.choices([False, True], weights=[0.85, 0.15])[0]

    if captain_cancelled:
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        reassigned_app = random.choice([a for a in selected_apps if a !=
→winning_app])
        reassign_delay_sec = random.randint(15, 60)
        total_ttf_sec = time_to_first_accept_sec + reassign_delay_sec + random.
→randint(10, 45)
        trip_status = random.choices(["Completed", "User_Cancelled"],
→weights=[0.88, 0.12])[0]
    else:
        reassigned_app = None
        total_ttf_sec = time_to_first_accept_sec
        trip_status = "Completed"

    data.append({
        "request_id": request_id,
        "user_id": user_id,
        "request_timestamp": request_time.strftime("%Y-%m-%d %H:%M:%S"),
        "connected_apps": connected_apps_str,
        "winning_app": winning_app,
        "time_to_first_accept_sec": time_to_first_accept_sec,
        "auto_cancel_status": auto_cancel_status,
        "captain_cancelled": captain_cancelled,
        "reassigned_app": reassigned_app,
        "total_time_to_fulfill_sec": total_ttf_sec,
        "trip_status": trip_status
    })

# Convert to DataFrame and export
df = pd.DataFrame(data)
df.to_csv("ride_multi_app_simulated_data.csv", index=False)
print("Successfully generated ride_multi_app_simulated_data.csv with 10,000
→records.")

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Successfully generated ride_multi_app_simulated_data.csv with 10,000 records.

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[2]: # Preview the first 5 rows of your simulated dataset
import pandas as pd

df = pd.read_csv("ride_multi_app_simulated_data.csv")
df.head()

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[2]:  request_id  user_id  request_timestamp  connected_apps \
0  REQ00001  USR4744  2026-08-15 01:24:00  Uber, Namma Yatri, Ola
1  REQ00002  USR4745  2026-08-22 23:44:00  Namma Yatri, Ola
2  REQ00003  USR2676  2026-08-23 03:44:00  Ola, Rapido, Namma Yatri, Uber
3  REQ00004  USR2439  2026-08-23 17:48:00  Ola, Rapido, Namma Yatri, Uber
4  REQ00005  USR3561  2026-08-12 02:26:00  Rapido, Namma Yatri

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	winning_app	time_to_first_accept_sec	auto_cancel_status	\
0	Uber	61	Success	
1	Namma Yatri	85	Success	
2	Namma Yatri	75	Success	
3	Namma Yatri	28	Success	
4	Namma Yatri	59	Delayed_Fee_Incurred	

	captain_cancelled	reassigned_app	total_time_to_fulfill_sec	trip_status
0	False	NaN	61	Completed
1	False	NaN	85	Completed
2	False	NaN	75	Completed
3	False	NaN	28	Completed
4	False	NaN	59	Completed

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[3]: df.info()
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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   request_id                            10000 non-null  object
1   user_id                               10000 non-null  object
2   request_timestamp                     10000 non-null  object
3   connected_apps                        10000 non-null  object
4   winning_app                           10000 non-null  object
5   time_to_first_accept_sec              10000 non-null  int64
6   auto_cancel_status                    10000 non-null  object
7   captain_cancelled                     10000 non-null  bool
8   reassigned_app                        1485 non-null   object
9   total_time_to_fulfill_sec             10000 non-null  int64
10  trip_status                           10000 non-null  object
dtypes: bool(1), int64(2), object(8)
memory usage: 791.1+ KB
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[4]: # Show the first 5 rows (default) or pass a number like df.head(10)
df.head()

# Show the last 5 rows
df.tail()

# Show 5 randomly sampled rows
df.sample(5)
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[4]:      request_id  user_id  request_timestamp  connected_apps \
2443  REQ02444  USR3865  2026-08-06 18:22:00  Uber, Ola, Rapido, Namma Yatri
6307  REQ06308  USR4696  2026-08-20 11:59:00  Namma Yatri, Rapido, Uber
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7207	REQ07208	USR1154	2026-08-01 18:39:00	Ola, Namma Yatri, Rapido, Uber
3355	REQ03356	USR2382	2026-08-14 06:29:00	Ola, Rapido
2435	REQ02436	USR1147	2026-08-14 05:20:00	Ola, Namma Yatri, Uber

	winning_app	time_to_first_accept_sec	auto_cancel_status	\
2443	Ola	35	Success	
6307	Rapido	25	Success	
7207	Namma Yatri	50	Success	
3355	Ola	34	Success	
2435	Namma Yatri	53	Success	

	captain_cancelled	reassigned_app	total_time_to_fulfill_sec	trip_status
2443	True	Rapido	128	Completed
6307	True	Namma Yatri	110	Completed
7207	False	NaN	50	Completed
3355	False	NaN	34	Completed
2435	False	NaN	53	Completed

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